

Домашнее задание N1. Установка и настройка Suricata

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Введение (цель работы)

Установка и настройка Suricata, создание и проверка трех собственных сигнатур, написание отчета.

Установка и базовая настройка Suricata

Подготовка

Работаем в виртуальной машине Ubuntu 24.04.5 LTS.

Смотрим имеющиеся интерфейсы:

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:c1:84:f1 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.179.129/24 brd 192.168.179.255 scope global dynamic noprefixroute ens33
        valid_lft 1420sec preferred_lft 1420sec
    inet6 fe80::20c:29ff:fec1:84f1/64 scope link
        valid_lft forever preferred_lft forever
3: br-9aae4843d23f: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 72:c3:42:11:1a:53 brd ff:ff:ff:ff:ff:ff
    inet 172.18.0.1/16 brd 172.18.255.255 scope global br-9aae4843d23f
        valid_lft forever preferred_lft forever
    inet6 fe80::70c3:42ff:fe11:1a53/64 scope link
        valid_lft forever preferred_lft forever
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 2a:b6:ef:fc:a3:2e brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
5: vethbc1f005@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-9aae4843d23f state UP group default
    link/ether e2:0d:d9:64:a9:8e brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet6 fe80::e00d:d9ff:fe64:a98e/64 scope link
        valid_lft forever preferred_lft forever
6: veth67e1fa3@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-9aae4843d23f state UP group default
    link/ether 0a:55:2e:18:fa:84 brd ff:ff:ff:ff:ff:ff link-netnsid 1
    inet6 fe80::109f:f8ff:fea4:221c/64 scope link
        valid_lft forever preferred_lft forever
7: veth7cc3ad8@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-9aae4843d23f state UP group default
    link/ether 26:55:64:97:00:50 brd ff:ff:ff:ff:ff:ff link-netnsid 2
    inet6 fe80::2455:64ff:fe97:50/64 scope link
        valid_lft forever preferred_lft forever
```

Понимаем, что основной ens33. На нем адрес 192.168.179.129/24.

Установка Suricata

Ставим Suricata из репозитория Ubuntu.

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo apt
install suricata
```

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ suricata -V
```

This is Suricata version 7.0.3 RELEASE

Подтверждение.

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ apt policy suricata
suricata -V
suricata:
  Installed: 1:7.0.3-1build3
  Candidate: 1:7.0.3-1build3
  Version table:
 *** 1:7.0.3-1build3 500
      500 http://am.archive.ubuntu.com/ubuntu noble/universe amd64 Packages
      100 /var/lib/dpkg/status
This is Suricata version 7.0.3 RELEASE
```

Настройка Suricata

Сразу не заработало.

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo systemctl status suricata --no-pager
```

```
× suricata.service - Suricata IDS/IDP daemon
```

```
Loaded: loaded (/usr/lib/systemd/system/suricata.service; enabled; preset: enabled)
```

```
Active: failed (Result: exit-code) since Wed 2026-09-09 10:44:25 +04; 45s ago
```

```
Duration: 58ms
```

```
Docs: man:suricata(8)
```

```
man:suricatasc(8)
```

```
https://suricata.io/documentation/
```

```
Process: 302474 ExecStart=/usr/bin/suricata -D --af-packet -c /etc/suricata/suricata.yaml --pidfile /run/suricata.pid (code=exited, status=0/SUCCESS)
```

```
Main PID: 302479 (code=exited, status=1/FAILURE)
```

```
CPU: 92ms
```

```
Sep 09 10:44:25 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: suricata.service: Scheduled restart job, restart counter is at 5.
```

```
Sep 09 10:44:25 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: suricata.service: Start request repeated too quickly.
```

```
Sep 09 10:44:25 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: suricata.service: Failed with result 'exit-code'.
```

```
Sep 09 10:44:25 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: Failed to start suricata.service - Suricata IDS/IDP daemon.
```

Разбираемся в причинах. Смотрим логи.

```

vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo journalctl -u suricata -n 50 --no-pager
Sep 09 10:44:23 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: Starting suricata.service - Suricata IDS/IDP daemon...
Sep 09 10:44:23 vladimir-kuryndin-VMware-Virtual-Platform suricata[302354]: i: suricata: This is Suricata version 7.0.3 RELEASE running in SYSTEM mode
Sep 09 10:44:23 vladimir-kuryndin-VMware-Virtual-Platform suricata[302354]: W: iocctl: Failure when trying to get MTU via iocctl for 'eth0': No such device (19)
Sep 09 10:44:23 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: Started suricata.service - Suricata IDS/IDP daemon.
Sep 09 10:44:23 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: suricata.service: Main process exited, code=exited, status=1/FAILURE
Sep 09 10:44:23 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: suricata.service: Failed with result 'exit-code'.
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: suricata.service: Scheduled restart job, restart counter is at 1.
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: Starting suricata.service - Suricata IDS/IDP daemon...
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform suricata[302380]: i: suricata: This is Suricata version 7.0.3 RELEASE running in SYSTEM mode
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform suricata[302380]: W: iocctl: Failure when trying to get MTU via iocctl for 'eth0': No such device (19)
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: Started suricata.service - Suricata IDS/IDP daemon.
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: suricata.service: Main process exited, code=exited, status=1/FAILURE
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: suricata.service: Failed with result 'exit-code'.
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: suricata.service: Scheduled restart job, restart counter is at 2.
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: Starting suricata.service - Suricata IDS/IDP daemon...
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform suricata[302405]: i: suricata: This is Suricata version 7.0.3 RELEASE running in SYSTEM mode
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform suricata[302405]: W: iocctl: Failure when trying to get MTU via iocctl for 'eth0': No such device (19)
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: Started suricata.service - Suricata IDS/IDP daemon.
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: suricata.service: Main process exited, code=exited, status=1/FAILURE
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: suricata.service: Failed with result 'exit-code'.
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: suricata.service: Scheduled restart job, restart counter is at 3.
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: Starting suricata.service - Suricata IDS/IDP daemon...
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform suricata[302425]: i: suricata: This is Suricata version 7.0.3 RELEASE running in SYSTEM mode
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform suricata[302425]: W: iocctl: Failure when trying to get MTU via iocctl for 'eth0': No such device (19)
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: Started suricata.service - Suricata IDS/IDP daemon.
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: suricata.service: Main process exited, code=exited, status=1/FAILURE
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: suricata.service: Failed with result 'exit-code'.
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: suricata.service: Scheduled restart job, restart counter is at 4.
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: Starting suricata.service - Suricata IDS/IDP daemon...
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform suricata[302474]: i: suricata: This is Suricata version 7.0.3 RELEASE running in SYSTEM mode
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform suricata[302474]: W: iocctl: Failure when trying to get MTU via iocctl for 'eth0': No such device (19)
Sep 09 10:44:24 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: Started suricata.service - Suricata IDS/IDP daemon.
Sep 09 10:44:25 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: suricata.service: Main process exited, code=exited, status=1/FAILURE
Sep 09 10:44:25 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: suricata.service: Failed with result 'exit-code'.
Sep 09 10:44:25 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: suricata.service: Scheduled restart job, restart counter is at 5.
Sep 09 10:44:25 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: suricata.service: Start request repeated too quickly.
Sep 09 10:44:25 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: suricata.service: Failed with result 'exit-code'.
Sep 09 10:44:25 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: Failed to start suricata.service - Suricata IDS/IDP daemon.

```

Понимаем, что suricata по умолчанию слушает интерфейс eth0. А у нас ens33.

Меняем интерфейс. Для этого сначала делаем бэкап файл конфигурации и далее правим файл с помощью nano

```

vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo cp
/etc/suricata/suricata.yaml /etc/suricata/suricata.yaml.bak

```

```

vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo nano
/etc/suricata/suricata.yaml

```

```
##
## Step 3: Configure common capture settings
##
## See "Advanced Capture Options" below for more options, including Netmap
## and PF_RING.
##

# Linux high speed capture support
af-packet:
- interface: ens33
  # Number of receive threads. "auto" uses the number of cores
  #threads: auto
  # Default clusterid. AF_PACKET will load balance packets based on flow.
  cluster-id: 99
  # Default AF_PACKET cluster type. AF_PACKET can load balance per flow or per hash.
  # This is only supported for Linux kernel > 3.1
  # possible value are:
  # * cluster_flow: all packets of a given flow are sent to the same socket
  # * cluster_cpu: all packets treated in kernel by a CPU are sent to the same socket
  # * cluster_qm: all packets linked by network card to a RSS queue are sent to the same
  # socket. Requires at least Linux 3.14.
  # * cluster_ebpf: eBPF file load balancing. See doc/userguide/capture-hardware/ebpf-xdp.rst for
  # more info.
  # Recommended modes are cluster_flow on most boxes and cluster_cpu or cluster_qm on system
  # with capture card using RSS (requires cpu affinity tuning and system IRQ tuning)
  # cluster_rollover has been deprecated; if used, it'll be replaced with cluster_flow.
  cluster-type: cluster_flow
  # In some fragmentation cases, the hash can not be computed. If "defrag" is set
  # to yes, the kernel will do the needed defragmentation before sending the packets.
  defrag: yes
  # To use the ring feature of AF_PACKET, set 'use-mmap' to yes
```

После этого:

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo suricata
-T -c /etc/suricata/suricata.yaml

i: suricata: This is Suricata version 7.0.3 RELEASE running in SYSTEM mode

W: detect: No rule files match the pattern
/var/lib/suricata/rules/suricata.rules

vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo systemctl
restart suricata

vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo systemctl
status suricata --no-pager
```

- suricata.service - Suricata IDS/IDP daemon

```
Loaded: loaded (/usr/lib/systemd/system/suricata.service; enabled;
preset: enabled)
```

```
Active: active (running) since Wed 2026-09-09 10:52:30 +04; 6s ago
```

```
Docs: man:suricata(8)
```

```
man:suricatasc(8)
```

```
https://suricata.io/documentation/
```

```
Process: 304851 ExecStart=/usr/bin/suricata -D --af-packet -c
/etc/suricata/suricata.yaml --pidfile /run/suricata.pid (code=exited,
status=0/SUCCESS)
```

```
Main PID: 304852 (Suricata-Main)
```

```
Tasks: 8 (limit: 7659)
```

```
Memory: 42.7M (peak: 43.1M)
```

```
CPU: 121ms
```

```
CGroup: /system.slice/suricata.service
```

```
└─304852 /usr/bin/suricata -D --af-packet -c
/etc/suricata/suricata.yaml --pidfile /run/suricata.pid
```

Создание и проверка трёх собственных сигнатур

Настройка файла local.rules

Выясним, где нам нужно создать правило.

```
ladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ grep -n
"default-rule-path" /etc/suricata/suricata.yaml
```

```
2150:default-rule-path: /var/lib/suricata/rules
```

Выясним, какую секцию файла suricata.yaml менять:

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ grep -n -A 10
"^rule-files:" /etc/suricata/suricata.yaml
```

```
2152:rule-files:
```

```
2153- - suricata.rules
```

```
2154-
```

```
2155-##
```

```
2156-## Auxiliary configuration files.
```

```
2157-##
```

```
2158-
```

```
2159-classification-file: /etc/suricata/classification.config
```

```
2160-reference-config-file: /etc/suricata/reference.config
```

```
2161-# threshold-file: /etc/suricata/threshold.config
```

```
2162-
```

Настраиваем, чтобы Suricata использовала local.rules.

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo touch
/var/lib/suricata/rules/local.rules
```

```
[sudo] password for vladimir-kuryndin:

vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ ls -l
/var/lib/suricata/rules/

total 0

-rw-r--r-- 1 root root 0 Sep  9 11:58 local.rules

vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo nano
/etc/suricata/suricata.yaml
```

```
##
## Configure Suricata to load Suricata-Update managed rules.
##

default-rule-path: /var/lib/suricata/rules

rule-files:
- local.rules

##
## Auxiliary configuration files.
##

classification-file: /etc/suricata/classification.config
reference-config-file: /etc/suricata/reference.config
# threshold-file: /etc/suricata/threshold.config
```

Перезапускаем и убеждаемся, что все работает

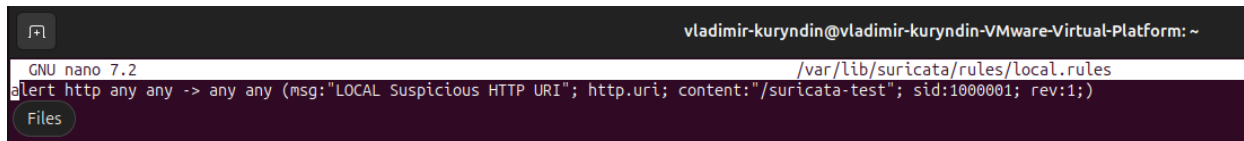
```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo suricata -T -c /etc/suricata/suricata.yaml
[sudo] password for vladimir-kuryndin:
GSorry, try again.
[sudo] password for vladimir-kuryndin:
i: suricata: This is Suricata version 7.0.3 RELEASE running in SYSTEM mode
W: detect: 1 rule files specified, but no rules were loaded!
i: suricata: Configuration provided was successfully loaded. Exiting.
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo suricata -T -c /etc/suricata/suricata.yaml
i: suricata: This is Suricata version 7.0.3 RELEASE running in SYSTEM mode
W: detect: 1 rule files specified, but no rules were loaded!
i: suricata: Configuration provided was successfully loaded. Exiting.
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ ^C
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo nano /var/lib/suricata/rules/local.rules
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo suricata -T -c /etc/suricata/suricata.yaml
i: suricata: This is Suricata version 7.0.3 RELEASE running in SYSTEM mode
i: suricata: Configuration provided was successfully loaded. Exiting.
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo systemctl restart suricata
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo systemctl status suricata --no-pager
● suricata.service - Suricata IDS/IDP daemon
   Loaded: loaded (/usr/lib/systemd/system/suricata.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-09-09 13:53:56 +04; 5s ago
     Docs: man:suricata(8)
           man:suricatasc(8)
           https://suricata.io/documentation/
  Process: 356594 ExecStart=/usr/bin/suricata -D --af-packet -c /etc/suricata/suricata.yaml --pidfile /run/suricata.pid (code=exited, status=0/SUCCESS)
 Main PID: 356596 (Suricata-Main)
    Tasks: 8 (limit: 7659)
   Memory: 42.9M (peak: 43.3M)
      CPU: 111ms
  CGroup: /system.slice/suricata.service
          └─356596 /usr/bin/suricata -D --af-packet -c /etc/suricata/suricata.yaml --pidfile /run/suricata.pid

Sep 09 13:53:56 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: Starting suricata.service - Suricata IDS/IDP daemon...
Sep 09 13:53:56 vladimir-kuryndin-VMware-Virtual-Platform suricata[356594]: i: suricata: This is Suricata version 7.0.3 RELEASE running in SYSTEM mode
Sep 09 13:53:56 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: Started suricata.service - Suricata IDS/IDP daemon.
```

Первое правило. HTTP-запрос

Сделаем первое правило

```
alert http any any -> any any (msg:"LOCAL Suspicious HTTP URI"; http.uri; content: "/suricata-test";  
sid:1000001; rev:1;)
```



```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform: ~  
GNU nano 7.2 /var/lib/suricata/rules/local.rules  
alert http any any -> any any (msg:"LOCAL Suspicious HTTP URI"; http.uri; content: "/suricata-test"; sid:1000001; rev:1;)  
Files
```

Это правило означает: если Suricata увидит HTTP-запрос, содержащий URI /suricata-test, она создаст alert с SID 1000001.

Проверяем:



```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform: ~$ curl http://example.com/suricata-test  
<!doctype html><html lang="en"><head><title>Example Domain</title><link rel="icon" href="data:,"><meta name="viewport" content="width=device-width, initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</style></head><body><div><h1>Example Domain</h1><p>This domain is for use in documentation examples without needing permission. Avoid use in operations.</p><p><a href="https://iana.org/domains/example">Learn more</a></p></div></body></html>  
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform: ~$ sudo tail -n 20 /var/log/suricata/fast.log  
09/09/2026-13:54:14.704257  [**] [1:1000001:1] LOCAL Suspicious HTTP URI [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.179.129:43236 -> 172.66.147.243:80
```

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform: ~$ curl  
http://example.com/suricata-test
```

```
<!doctype html><html lang="en"><head><title>Example Domain</title><link  
rel="icon" href="data:,"><meta name="viewport" content="width=device-width,  
initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh  
auto;font-family:system-ui,sans-serif}h1{font-  
size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</style></head><body><  
div><h1>Example Domain</h1><p>This domain is for use in documentation  
examples without needing permission. Avoid use in operations.</p><p><a  
href="https://iana.org/domains/example">Learn  
more</a></p></div></body></html>
```

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform: ~$ sudo tail -n  
20 /var/log/suricata/fast.log
```

```
09/09/2026-13:54:14.704257  [**] [1:1000001:1] LOCAL Suspicious HTTP URI [**]  
[Classification: (null)] [Priority: 3] {TCP} 192.168.179.129:43236 ->  
172.66.147.243:80
```

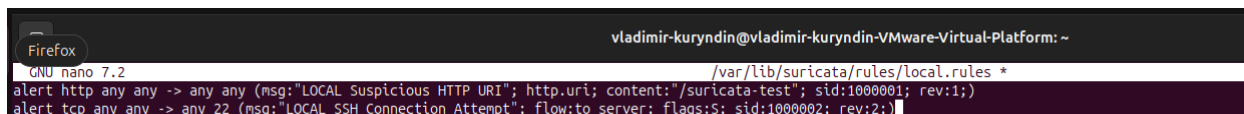
Первое правило успешно сработало.

Второе правило. Обнаружение попытки SSH-подключения

Делаем следующее правило

```
alert tcp any any -> any 22 (msg:"LOCAL SSH Connection Attempt";  
flow:to_server; flags:S; sid:1000002; rev:2;)
```

Правило делает следующее: отслеживает TCP-пакеты, направленные на **порт 22**. Параметр flags:S означает, что правило реагирует на TCP SYN-пакет — начало попытки установить соединение.



```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform: ~  
GNU nano 7.2 /var/lib/suricata/rules/local.rules *  
alert http any any -> any any (msg:"LOCAL Suspicious HTTP URI"; http.uri; content: "/suricata-test"; sid:1000001; rev:1;)  
alert tcp any any -> any 22 (msg:"LOCAL SSH Connection Attempt"; flow:to_server; flags:S; sid:1000002; rev:2;)
```

Проверяем:

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ ssh -o  
ConnectTimeout=5 test@example.com
```

```
ssh: connect to host example.com port 22: Network is unreachable
```

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo tail -n  
20 /var/log/suricata/fast.log
```

```
09/09/2026-13:54:14.704257  [**] [1:1000001:1] LOCAL Suspicious HTTP URI [**]  
[Classification: (null)] [Priority: 3] {TCP} 192.168.179.129:43236 ->  
172.66.147.243:80
```

```
09/09/2026-14:05:44.277115  [**] [1:1000002:1] LOCAL SSH Connection Attempt  
[**] [Classification: (null)] [Priority: 3] {TCP} 192.168.179.129:36664 ->  
104.20.23.154:22
```

```
09/09/2026-14:05:45.291470  [**] [1:1000002:1] LOCAL SSH Connection Attempt  
[**] [Classification: (null)] [Priority: 3] {TCP} 192.168.179.129:36664 ->  
104.20.23.154:22
```

```
09/09/2026-14:05:46.315172  [**] [1:1000002:1] LOCAL SSH Connection Attempt  
[**] [Classification: (null)] [Priority: 3] {TCP} 192.168.179.129:36664 ->  
104.20.23.154:22
```

```
09/09/2026-14:05:47.339416  [**] [1:1000002:1] LOCAL SSH Connection Attempt  
[**] [Classification: (null)] [Priority: 3] {TCP} 192.168.179.129:36664 ->  
104.20.23.154:22
```

```
09/09/2026-14:05:48.363126  [**] [1:1000002:1] LOCAL SSH Connection Attempt  
[**] [Classification: (null)] [Priority: 3] {TCP} 192.168.179.129:36664 ->  
104.20.23.154:22
```

```
09/09/2026-14:05:49.280842  [**] [1:1000002:1] LOCAL SSH Connection Attempt  
[**] [Classification: (null)] [Priority: 3] {TCP} 192.168.179.129:49714 ->  
172.66.147.243:22
```

```
09/09/2026-14:05:50.283153  [**] [1:1000002:1] LOCAL SSH Connection Attempt  
[**] [Classification: (null)] [Priority: 3] {TCP} 192.168.179.129:49714 ->  
172.66.147.243:22
```

```
09/09/2026-14:05:51.307448  [**] [1:1000002:1] LOCAL SSH Connection Attempt  
[**] [Classification: (null)] [Priority: 3] {TCP} 192.168.179.129:49714 ->  
172.66.147.243:22
```

```
09/09/2026-14:05:52.331132  [**] [1:1000002:1] LOCAL SSH Connection Attempt  
[**] [Classification: (null)] [Priority: 3] {TCP} 192.168.179.129:49714 ->  
172.66.147.243:22
```

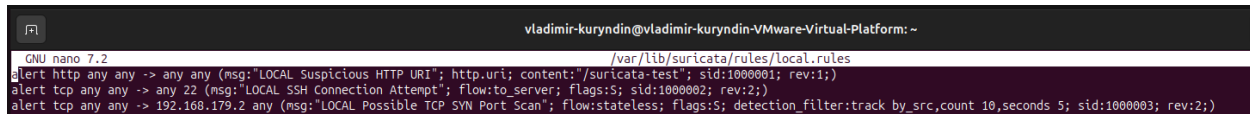
```
09/09/2026-14:05:53.355343  [**] [1:1000002:1] LOCAL SSH Connection Attempt  
[**] [Classification: (null)] [Priority: 3] {TCP} 192.168.179.129:49714 ->  
172.66.147.243:22
```

Второе правило успешно сработало.

Третье правило. Обнаружение TCP-сканирования портов через nmap

Делаем следующее правило:

```
alert tcp any any -> 192.168.179.2 any (msg:"LOCAL Possible TCP SYN Port Scan"; flow:stateless; flags:S; detection_filter:track by_src,count 10,seconds 5; sid:1000003; rev:2;)
```



The screenshot shows a terminal window with the title 'vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform: ~'. The nano editor is open, editing the file '/var/lib/suricata/rules/local.rules'. The content of the file is as follows:

```
GNU nano 7.2 /var/lib/suricata/rules/local.rules
alert http any any -> any any (msg:"LOCAL Suspicious HTTP URI"; http.uri; content:"/suricata-test"; sid:1000001; rev:1;)
alert tcp any any -> any 22 (msg:"LOCAL SSH Connection Attempt"; flow:to_server; flags:S; sid:1000002; rev:2;)
alert tcp any any -> 192.168.179.2 any (msg:"LOCAL Possible TCP SYN Port Scan"; flow:stateless; flags:S; detection_filter:track by_src,count 10,seconds 5; sid:1000003; rev:2;)
```

Проверяем:

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo nmap -sS -p 1-100 192.168.179.2
```

Starting Nmap 7.94SVN (<https://nmap.org>) at 2026-09-09 15:29 +04

Nmap scan report for _gateway (192.168.179.2)

Host is up (0.00091s latency).

Not shown: 99 closed tcp ports (reset)

PORT	STATE	SERVICE
------	-------	---------

53/tcp	open	domain
--------	------	--------

MAC Address: 00:50:56:E7:CF:03 (VMware)

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo jq -c 'select(.event_type=="alert" and .alert.signature_id==1000003) | {timestamp,src_ip,dest_ip,dest_port,action:.alert.action,signature:.alert.signature}' /var/log/suricata/eve.json | tail -n 20
```

```
{
  "timestamp": "2026-09-09T15:29:25.513419+0400",
  "src_ip": "192.168.179.129",
  "dest_ip": "192.168.179.2",
  "dest_port": 79,
  "action": "allowed",
  "signature": "LOCAL Possible TCP SYN Port Scan"
}
```

```
{
  "timestamp": "2026-09-09T15:29:25.513450+0400",
  "src_ip": "192.168.179.129",
  "dest_ip": "192.168.179.2",
  "dest_port": 57,
  "action": "allowed",
  "signature": "LOCAL Possible TCP SYN Port Scan"
}
```

```
{
  "timestamp": "2026-09-09T15:29:25.513474+0400",
  "src_ip": "192.168.179.129",
  "dest_ip": "192.168.179.2",
  "dest_port": 34,
  "action": "allowed",
  "signature": "LOCAL Possible TCP SYN Port Scan"
}
```

```
{
  "timestamp": "2026-09-09T15:29:25.513496+0400",
  "src_ip": "192.168.179.129",
  "dest_ip": "192.168.179.2",
  "dest_port": 3,
  "action": "allowed",
  "signature": "LOCAL Possible TCP SYN Port Scan"
}
```

```
{"timestamp":"2026-09-09T15:29:25.513499+0400","src_ip":"192.168.179.129","dest_ip":"192.168.179.2","dest_port":88,"action":"allowed","signature":"LOCAL Possible TCP SYN Port Scan"}
```

```
{"timestamp":"2026-09-09T15:29:25.513499+0400","src_ip":"192.168.179.129","dest_ip":"192.168.179.2","dest_port":43,"action":"allowed","signature":"LOCAL Possible TCP SYN Port Scan"}
```

```
{"timestamp":"2026-09-09T15:29:25.513500+0400","src_ip":"192.168.179.129","dest_ip":"192.168.179.2","dest_port":86,"action":"allowed","signature":"LOCAL Possible TCP SYN Port Scan"}
```

```
{"timestamp":"2026-09-09T15:29:25.513536+0400","src_ip":"192.168.179.129","dest_ip":"192.168.179.2","dest_port":83,"action":"allowed","signature":"LOCAL Possible TCP SYN Port Scan"}
```

```
{"timestamp":"2026-09-09T15:29:25.513536+0400","src_ip":"192.168.179.129","dest_ip":"192.168.179.2","dest_port":19,"action":"allowed","signature":"LOCAL Possible TCP SYN Port Scan"}
```

```
{"timestamp":"2026-09-09T15:29:25.513537+0400","src_ip":"192.168.179.129","dest_ip":"192.168.179.2","dest_port":29,"action":"allowed","signature":"LOCAL Possible TCP SYN Port Scan"}
```

```
{"timestamp":"2026-09-09T15:29:25.513539+0400","src_ip":"192.168.179.129","dest_ip":"192.168.179.2","dest_port":48,"action":"allowed","signature":"LOCAL Possible TCP SYN Port Scan"}
```

```
{"timestamp":"2026-09-09T15:29:25.513986+0400","src_ip":"192.168.179.129","dest_ip":"192.168.179.2","dest_port":37,"action":"allowed","signature":"LOCAL Possible TCP SYN Port Scan"}
```

```
{"timestamp":"2026-09-09T15:29:25.513989+0400","src_ip":"192.168.179.129","dest_ip":"192.168.179.2","dest_port":66,"action":"allowed","signature":"LOCAL Possible TCP SYN Port Scan"}
```

```
{"timestamp":"2026-09-09T15:29:25.513990+0400","src_ip":"192.168.179.129","dest_ip":"192.168.179.2","dest_port":33,"action":"allowed","signature":"LOCAL Possible TCP SYN Port Scan"}
```

```
{"timestamp":"2026-09-09T15:29:25.513991+0400","src_ip":"192.168.179.129","dest_ip":"192.168.179.2","dest_port":98,"action":"allowed","signature":"LOCAL Possible TCP SYN Port Scan"}
```

```
{ "timestamp": "2026-09-09T15:29:25.513982+0400", "src_ip": "192.168.179.129", "dest_ip": "192.168.179.2", "dest_port": 46, "action": "allowed", "signature": "LOCAL Possible TCP SYN Port Scan" }

{ "timestamp": "2026-09-09T15:29:25.513986+0400", "src_ip": "192.168.179.129", "dest_ip": "192.168.179.2", "dest_port": 12, "action": "allowed", "signature": "LOCAL Possible TCP SYN Port Scan" }

{ "timestamp": "2026-09-09T15:29:25.513988+0400", "src_ip": "192.168.179.129", "dest_ip": "192.168.179.2", "dest_port": 40, "action": "allowed", "signature": "LOCAL Possible TCP SYN Port Scan" }

{ "timestamp": "2026-09-09T15:29:25.513990+0400", "src_ip": "192.168.179.129", "dest_ip": "192.168.179.2", "dest_port": 65, "action": "allowed", "signature": "LOCAL Possible TCP SYN Port Scan" }

{ "timestamp": "2026-09-09T15:29:25.513991+0400", "src_ip": "192.168.179.129", "dest_ip": "192.168.179.2", "dest_port": 32, "action": "allowed", "signature": "LOCAL Possible TCP SYN Port Scan" }
```

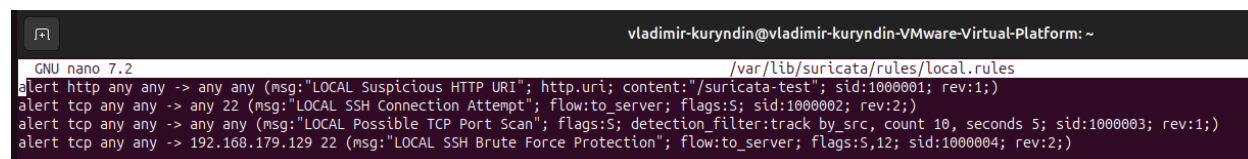
Третье правило успешно сработало.

Четвертое правило. Временная блокировка новых SSH-подключений при brute force
Сделаем следующее правило: имитируем SSH brute force серией частых подключений и временно заблокируем новые SSH-подключения от источника, генерирующего слишком много попыток соединения.

Создаем правило local.rules

Создаем правило:

```
alert tcp any any -> 192.168.179.129 22 (msg:"LOCAL SSH Brute Force Protection"; flow:to_server; flags:S,12; sid:1000004; rev:2;)
```



```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform: ~
GNU nano 7.2 /var/lib/suricata/rules/local.rules
alert http any any -> any any (msg:"LOCAL Suspicious HTTP URI"; http.uri; content:"/suricata-test"; sid:1000001; rev:1;)
alert tcp any any -> any 22 (msg:"LOCAL SSH Connection Attempt"; flow:to_server; flags:S; sid:1000002; rev:2;)
alert tcp any any -> any any (msg:"LOCAL Possible TCP Port Scan"; flags:S; detection_filter:track by_src, count 10, seconds 5; sid:1000003; rev:1;)
alert tcp any any -> 192.168.179.129 22 (msg:"LOCAL SSH Brute Force Protection"; flow:to_server; flags:S,12; sid:1000004; rev:2;)
```

Настраиваем rate_filter

Сделали соответствующий фильтр:

```
rate_filter gen_id 1, sig_id 1000004, track by_src, count 5, seconds 60,
new_action drop, timeout 300
```

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform: ~  
GNU nano 7.2 /etc/suricata/threshold.config  
# Thresholding:  
#  
# This feature is used to reduce the number of logged alerts for noisy rules.  
# Thresholding commands limit the number of times a particular event is logged  
# during a specified time interval.  
#  
# The syntax is the following:  
#  
# threshold gen_id <gen_id>, sig_id <sig_id>, type <limit|threshold|both>, track <by_src|by_dst>, count <n>, seconds <t>  
#  
# event_filter gen_id <gen_id>, sig_id <sig_id>, type <limit|threshold|both>, track <by_src|by_dst>, count <n>, seconds <t>  
#  
# suppress gen_id <gid>, sig_id <sid>  
# suppress gen_id <gid>, sig_id <sid>, track <by_src|by_dst>, ip <ip|subnet>  
#  
# The options are documented at https://docs.suricata.io/en/latest/configuration/global-thresholds.html  
#  
# Please note that thresholding can also be set inside a signature. The interaction between rule based thresholds  
# and global thresholds is documented here:  
# https://docs.suricata.io/en/latest/configuration/global-thresholds.html#global-thresholds-vs-rule-thresholds  
#  
# Limit to 10 alerts every 10 seconds for each source host  
#threshold gen_id 0, sig_id 0, type limit, track by_src, count 10, seconds 10  
#  
# Limit to 1 alert every 10 seconds for signature with sid 2404000 per destination host  
#threshold gen_id 1, sig_id 2404000, type limit, track by_dst, count 1, seconds 10  
#  
# Avoid to alert on f-secure update  
# Example taken from https://blog.inliniac.net/2012/03/07/f-secure-av-updates-and-suricata-ips/  
#suppress gen_id 1, sig_id 2009557, track by_src, ip 217.110.97.128/25  
#suppress gen_id 1, sig_id 2012086, track by_src, ip 217.110.97.128/25  
#suppress gen_id 1, sig_id 2003614, track by_src, ip 217.110.97.128/25  
rate_filter gen_id 1, sig_id 1000004, track by_src, count 5, seconds 60, new_action drop, timeout 300
```

Подключаем threshold.config в suricata.yaml

```
##  
## Auxiliary configuration files.  
##  
  
classification-file: /etc/suricata/classification.config  
reference-config-file: /etc/suricata/reference.config  
threshold-file: /etc/suricata/threshold.config  
  
##  
## Include other configs  
##  
  
# Includes: Files included here will be handled as if they were in-lined  
# in this configuration file. Files with relative pathnames will be  
# searched for in the same directory as this configuration file. You may  
# use absolute pathnames too.  
#include:  
# - include1.yaml  
# - include2.yaml
```

Переключаем Suricata в режим IPS/NFQUEUE

Далее переключаем нашу Suricata в режим IPS/NFQUEUE

Для этого делаем system override

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ cat  
/etc/systemd/system/suricata.service.d/override.conf
```

[Service]

ExecStart=

```
ExecStart=/usr/bin/suricata -D -q 0 -c /etc/suricata/suricata.yaml --pidfile /run/suricata.pid
```

Перезапускаем suricata

```
sudo systemctl daemon-reload
```

```
sudo systemctl restart suricata
```

и убеждаемся, что наша suricata работает в правильном режиме

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform: $ sudo systemctl status suricata --no-pager
● suricata.service - Suricata IDS/IDP daemon
   Loaded: loaded (/usr/lib/systemd/system/suricata.service; enabled; preset: enabled)
  Drop-In: /etc/systemd/system/suricata.service.d
           └─override.conf
   Active: active (running) since Wed 2026-09-09 14:39:40 +04; 8min ago
     Docs: man:suricata(8)
           man:suricata-sc(8)
           https://suricata.io/documentation/
  Process: 370876 ExecStart=/usr/bin/suricata -D -q 0 -c /etc/suricata/suricata.yaml --pidfile /run/suricata.pid (code=exited, status=0/SUCCESS)
    Main PID: 370877 (Suricata-Main)
      Tasks: 10 (limit: 7659)
    Memory: 48.8M (peak: 49.1M)
       CPU: 1.705s
    CGroup: /system.slice/suricata.service
            └─370877 /usr/bin/suricata -D -q 0 -c /etc/suricata/suricata.yaml --pidfile /run/suricata.pid

Sep 09 14:39:40 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: Starting suricata.service - Suricata IDS/IDP daemon...
Sep 09 14:39:40 vladimir-kuryndin-VMware-Virtual-Platform suricata[370876]: i: suricata: This is Suricata version 7.0.3 RELEASE running in SYSTEM mode
Sep 09 14:39:40 vladimir-kuryndin-VMware-Virtual-Platform systemd[1]: Started suricata.service - Suricata IDS/IDP daemon.
```

Настраиваем iptables

Далее настраиваем iptables. Добавляем следующее правило:

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo iptables
-I INPUT 1 -i ens33 -p tcp --dport 22 -j NFQUEUE --queue-num 0 --queue-bypass
```

Проверяем правило firewall

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo iptables
-vnL INPUT --line-numbers
```

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)

num	pkts	bytes	target	prot	opt	in	out	source	destination
1	0	0	NFQUEUE	6	--	ens33	*	0.0.0.0/0	0.0.0.0/0
									tcp dpt:22 NFQUEUE num 0 bypass

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo iptables
-vnL INPUT --line-numbers
```

Далее с другой машины пробуем подключиться по ssh:

```
vladimirkuryndin@fedora:~$ ssh vladimir-kuryndin@192.168.179.129
```

The authenticity of host '192.168.179.129 (192.168.179.129)' can't be established.

ED25519 key fingerprint is
SHA256:CnIt8B8Xyxdg3yL/iwXgvgLrtR3GilekIODe/8JmjZ0.

This key is not known by any other names.

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '192.168.179.129' (ED25519) to the list of known
hosts.
vladimir-kuryndin@192.168.179.129's password:
Permission denied, please try again.
vladimir-kuryndin@192.168.179.129's password:
Welcome to Ubuntu 24.04.5 LTS (GNU/Linux 7.0.0-31-generic x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro
```

Expanded Security Maintenance for Applications is not enabled.

4 updates can be applied immediately.

To see these additional updates run: `apt list --upgradable`

15 additional security updates can be applied with ESM Apps.

Learn more about enabling ESM Apps service at <https://ubuntu.com/esm>

vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~\$

Фиксируем, что firewall отработал

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo iptables
-vnL INPUT --line-numbers
```

Chain INPUT (policy ACCEPT 46 packets, 29194 bytes)

num	pkts	bytes	target	prot	opt	in	out	source
destination								
1	48	6941	NFQUEUE	6	--	ens33	*	0.0.0.0/0
			0.0.0.0/0	tcp	dpt:22	NFQUEUE	num 0	bypass

Проверка прошла успешно

Имитация brute-force и проверка блокировки

Далее на машине с Fedora запускаем:

```
vladimirkuryndin@fedora:~$ for i in {1..8}; do
    echo "Attempt $i"
    ssh -o BatchMode=yes \
        -o PubkeyAuthentication=no \
        -o PasswordAuthentication=no \
        -o KbdInteractiveAuthentication=no \
        -o ConnectTimeout=2 \
        vladimir-kuryndin@192.168.179.129 exit
done
Attempt 1
vladimir-kuryndin@192.168.179.129: Permission denied (publickey,password).
Attempt 2
vladimir-kuryndin@192.168.179.129: Permission denied (publickey,password).
Attempt 3
vladimir-kuryndin@192.168.179.129: Permission denied (publickey,password).
Attempt 4
vladimir-kuryndin@192.168.179.129: Permission denied (publickey,password).
Attempt 5
ssh: connect to host 192.168.179.129 port 22: Connection timed out
Attempt 6
ssh: connect to host 192.168.179.129 port 22: Connection timed out
Attempt 7
ssh: connect to host 192.168.179.129 port 22: Connection timed out
Attempt 8
ssh: connect to host 192.168.179.129 port 22: Connection timed out
vladimirkuryndin@fedora:~$
```

Замечание. Перед запуском серии уже выполнялось тестовое SSH-подключение, которое попало в 60-секундное окно `rate_filter`. Поэтому в данном запуске порог был достигнут на пятой попытке. При повторной проверке после истечения временного окна первые пять соединений были разрешены, а блокировка началась с шестого

Видим в логах на основном Ubuntu:

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo jq -c
'select(.event_type=="alert" and .alert.signature_id==1000004) |
```

```
{timestamp,src_ip,dest_ip,action:.alert.action,signature:.alert.signature}'  
/var/log/suricata/eve.json | tail -n 20
```

```
{"timestamp":"2026-09-09T14:52:43.585179+0400","src_ip":"192.168.179.130","dest_ip":"192.168.179.129","action":"allowed","signature":"LOCAL SSH Brute Force Protection"}
```

```
{"timestamp":"2026-09-09T14:59:24.966965+0400","src_ip":"192.168.179.130","dest_ip":"192.168.179.129","action":"allowed","signature":"LOCAL SSH Brute Force Protection"}
```

```
{"timestamp":"2026-09-09T15:00:08.769686+0400","src_ip":"192.168.179.130","dest_ip":"192.168.179.129","action":"allowed","signature":"LOCAL SSH Brute Force Protection"}
```

```
{"timestamp":"2026-09-09T15:00:08.837808+0400","src_ip":"192.168.179.130","dest_ip":"192.168.179.129","action":"allowed","signature":"LOCAL SSH Brute Force Protection"}
```

```
{"timestamp":"2026-09-09T15:00:08.902858+0400","src_ip":"192.168.179.130","dest_ip":"192.168.179.129","action":"allowed","signature":"LOCAL SSH Brute Force Protection"}
```

```
{"timestamp":"2026-09-09T15:00:08.968037+0400","src_ip":"192.168.179.130","dest_ip":"192.168.179.129","action":"allowed","signature":"LOCAL SSH Brute Force Protection"}
```

```
{"timestamp":"2026-09-09T15:00:09.034516+0400","src_ip":"192.168.179.130","dest_ip":"192.168.179.129","action":"blocked","signature":"LOCAL SSH Brute Force Protection"}
```

```
{"timestamp":"2026-09-09T15:00:10.073878+0400","src_ip":"192.168.179.130","dest_ip":"192.168.179.129","action":"blocked","signature":"LOCAL SSH Brute Force Protection"}
```

```
{"timestamp":"2026-09-09T15:00:11.041641+0400","src_ip":"192.168.179.130","dest_ip":"192.168.179.129","action":"blocked","signature":"LOCAL SSH Brute Force Protection"}
```

```
{"timestamp":"2026-09-09T15:00:12.058234+0400","src_ip":"192.168.179.130","dest_ip":"192.168.179.129","action":"blocked","signature":"LOCAL SSH Brute Force Protection"}
```

```
{"timestamp":"2026-09-09T15:00:13.051111+0400","src_ip":"192.168.179.130","dest_ip":"192.168.179.129","action":"blocked","signature":"LOCAL SSH Brute Force Protection"}
```

```
{"timestamp":"2026-09-09T15:00:14.106277+0400","src_ip":"192.168.179.130","dest_ip":"192.168.179.129","action":"blocked","signature":"LOCAL SSH Brute Force Protection"}
```

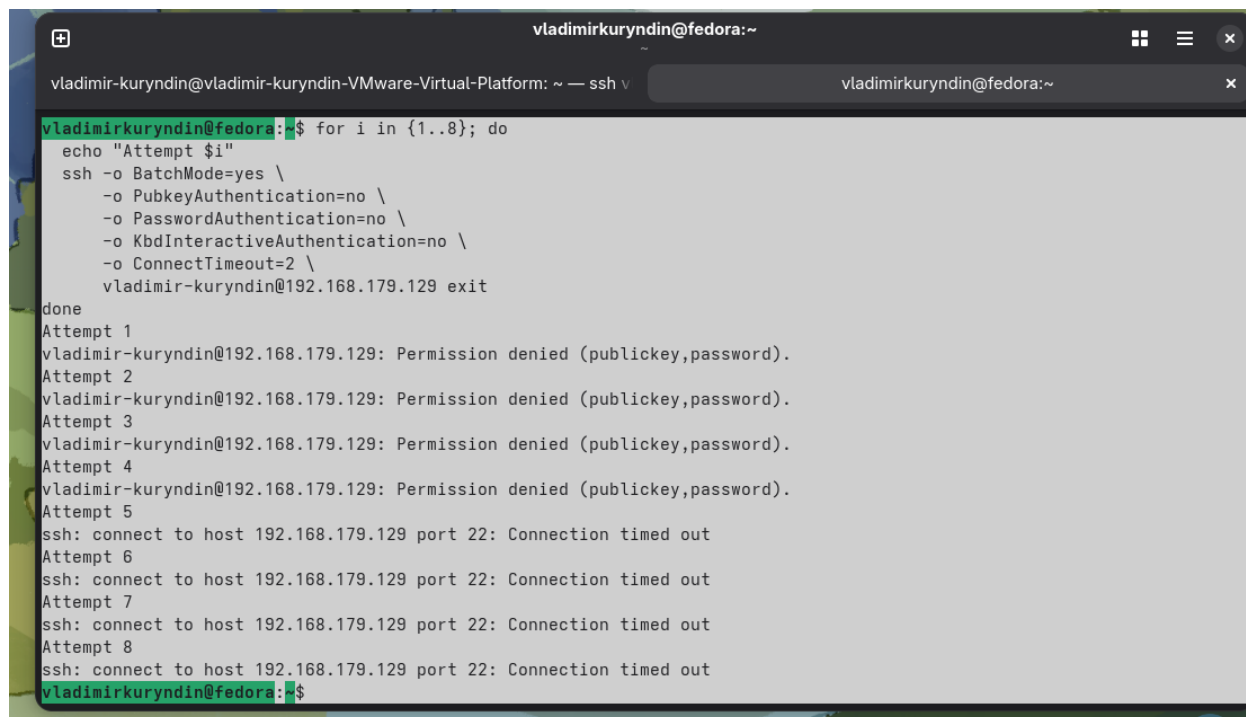
```
{"timestamp":"2026-09-09T15:00:15.057028+0400","src_ip":"192.168.179.130","dest_ip":"192.168.179.129","action":"blocked","signature":"LOCAL SSH Brute Force Protection"}
```

```
{ "timestamp": "2026-09-09T15:00:16.090152+0400", "src_ip": "192.168.179.130", "dest_ip": "192.168.179.129", "action": "blocked", "signature": "LOCAL SSH Brute Force Protection" }
```

vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~\$

Четвертое правило успешно сработало

Дополнительные скриншоты для фиксации:

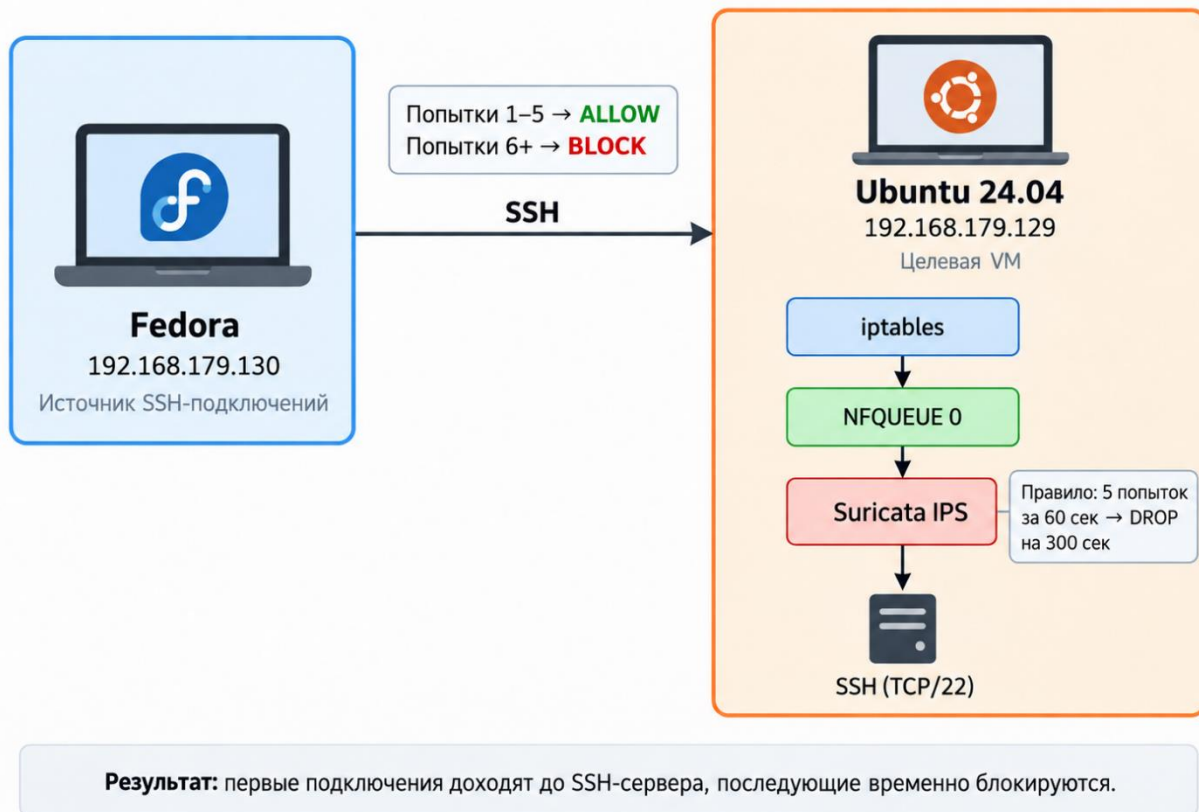


```
vladimirkuryndin@fedora:~$ for i in {1..8}; do
  echo "Attempt $i"
  ssh -o BatchMode=yes \
    -o PubkeyAuthentication=no \
    -o PasswordAuthentication=no \
    -o KbdInteractiveAuthentication=no \
    -o ConnectTimeout=2 \
    vladimir-kuryndin@192.168.179.129 exit
done
Attempt 1
vladimir-kuryndin@192.168.179.129: Permission denied (publickey,password).
Attempt 2
vladimir-kuryndin@192.168.179.129: Permission denied (publickey,password).
Attempt 3
vladimir-kuryndin@192.168.179.129: Permission denied (publickey,password).
Attempt 4
vladimir-kuryndin@192.168.179.129: Permission denied (publickey,password).
Attempt 5
ssh: connect to host 192.168.179.129 port 22: Connection timed out
Attempt 6
ssh: connect to host 192.168.179.129 port 22: Connection timed out
Attempt 7
ssh: connect to host 192.168.179.129 port 22: Connection timed out
Attempt 8
ssh: connect to host 192.168.179.129 port 22: Connection timed out
vladimirkuryndin@fedora:~$
```

```
vladimir-kuryndin@vladimir-kuryndin-VMware-Virtual-Platform:~$ sudo jq -c 'select(.event_type=="alert" and .alert.signature_id==1000004) | {timestamp,src_ip,dest_ip,action:.alert.action,signature:.alert.signature}' /var/log/suricata/eve.json | tail -n 20
{"timestamp": "2026-09-09T14:52:43.585179+0400", "src_ip": "192.168.179.130", "dest_ip": "192.168.179.129", "action": "allowed", "signature": "LOCAL SSH Brute Force Protection"}
{"timestamp": "2026-09-09T14:59:24.966965+0400", "src_ip": "192.168.179.130", "dest_ip": "192.168.179.129", "action": "allowed", "signature": "LOCAL SSH Brute Force Protection"}
{"timestamp": "2026-09-09T15:00:08.769686+0400", "src_ip": "192.168.179.130", "dest_ip": "192.168.179.129", "action": "allowed", "signature": "LOCAL SSH Brute Force Protection"}
{"timestamp": "2026-09-09T15:00:08.837088+0400", "src_ip": "192.168.179.130", "dest_ip": "192.168.179.129", "action": "allowed", "signature": "LOCAL SSH Brute Force Protection"}
{"timestamp": "2026-09-09T15:00:08.902858+0400", "src_ip": "192.168.179.130", "dest_ip": "192.168.179.129", "action": "allowed", "signature": "LOCAL SSH Brute Force Protection"}
{"timestamp": "2026-09-09T15:00:08.968837+0400", "src_ip": "192.168.179.130", "dest_ip": "192.168.179.129", "action": "allowed", "signature": "LOCAL SSH Brute Force Protection"}
{"timestamp": "2026-09-09T15:00:09.034516+0400", "src_ip": "192.168.179.130", "dest_ip": "192.168.179.129", "action": "blocked", "signature": "LOCAL SSH Brute Force Protection"}
{"timestamp": "2026-09-09T15:00:10.073878+0400", "src_ip": "192.168.179.130", "dest_ip": "192.168.179.129", "action": "blocked", "signature": "LOCAL SSH Brute Force Protection"}
{"timestamp": "2026-09-09T15:00:11.041641+0400", "src_ip": "192.168.179.130", "dest_ip": "192.168.179.129", "action": "blocked", "signature": "LOCAL SSH Brute Force Protection"}
{"timestamp": "2026-09-09T15:00:12.048923+0400", "src_ip": "192.168.179.130", "dest_ip": "192.168.179.129", "action": "blocked", "signature": "LOCAL SSH Brute Force Protection"}
{"timestamp": "2026-09-09T15:00:13.051111+0400", "src_ip": "192.168.179.130", "dest_ip": "192.168.179.129", "action": "blocked", "signature": "LOCAL SSH Brute Force Protection"}
{"timestamp": "2026-09-09T15:00:14.106277+0400", "src_ip": "192.168.179.130", "dest_ip": "192.168.179.129", "action": "blocked", "signature": "LOCAL SSH Brute Force Protection"}
{"timestamp": "2026-09-09T15:00:15.057028+0400", "src_ip": "192.168.179.130", "dest_ip": "192.168.179.129", "action": "blocked", "signature": "LOCAL SSH Brute Force Protection"}
{"timestamp": "2026-09-09T15:00:16.090152+0400", "src_ip": "192.168.179.130", "dest_ip": "192.168.179.129", "action": "blocked", "signature": "LOCAL SSH Brute Force Protection" }
```

Итоговый результат

В результате получили следующую картинку:



Выводы и сложности

В ходе работы мы настроили систему обнаружения и предотвращения сетевых атак на базе Suricata 7.0.3 под Ubuntu 24.04 LTS. Основной сетевой интерфейс — ens33, IP-адрес виртуальной машины — 192.168.179.129.

Разработали и успешно протестировали собственные правила для обнаружения HTTP-запроса, SSH-подключения, TCP-сканирования портов и SSH brute force. Для проверки использовали curl, ssh и Nmap 7.94SVN. Срабатывания подтверждены журналами fast.log и eve.json.

Дополнительно Suricata перевели из IDS-режима AF_PACKET в IPS-режим NFQUEUE. С помощью iptables 1.8.10 (nf_tables) реализовали передачу входящего SSH-трафика в Suricata. При достижении пяти SSH-подключений за 60 секунд последующие SSH-подключения от этого источника блокировались в течение 300 секунд.

Основные сложности были связаны с неверным интерфейсом eth0 в исходной конфигурации, подключением файла local.rules и настройкой NFQUEUE. После изменения интерфейса на ens33 и проверки конфигурации командой suricata -T все правила и IPS-блокировка работали корректно.